SG-345-300

الصناعات الدقيقة

PRECISION INDUSTRIES ®

Power Generator Packagers And Canopy Builders

GENSET NO LOAD – LOAD & PRE-TEST CHECKING REPORT

Date :		20 / JULY / 2017	
PI Model :	PI 313C	Type :	CLOSE
Prime Power :	284 KVA	Canopy Color :	ORANGE
Frq. : 60 Hz	$P.F_{LOAD}$ 1	Voltage :	380 V
Serial No :			
Job No :		8650	

CLIENT : SAUDI ARBIAN SERVICE SAR A/C

PROJECT :

GENSET DETAILS :						
Engine :	CUMMINS		Model No :	QSL9 – G3	Serial No :	84507971
Alternator :	STAMFORD		Model No :	UCDI 1274K1	Serial No :	N17D166242
Controller :	DEEP SEA		Model No :	6120	Serial No :	6419708
C. Breaker :	METASOL Amps	400 A	Model No :	ABN 803C	Serial No :	N/A
C. Transformer :	ELEMEX Amps	800/5 A	Model No :	N/A	Serial No :	N/A
AVR :	STAMFORD		Model No :	SX460	Serial No :	111505 - 2029

GENSET RATING :			
Testing Full Load Current :	345 Amps	Nominal Current :	432 Amps

LOAD TEST WILL BE IMPLEMENTED AL AMIENT CONDITIONS			
Temperature :	42^{0} C	Altitude :	

LOAD TEST

LOAD	RUN TIME	FREQ	SPEED	VOLTAGE L-L			CURRENT			POWER	COOLANT TEMP.	OIL PRESSURE	CHARGING ALTERNATOR
				L1	L2	L3	L1	L2	L3				
%	Min	Hz	RPM	V	V	V	A	A	A	KW	^{0}C	BAR	VDC
0%	5	60.0	1800	380	380	380	--	--	--	--	72	4.50	26.5
25%	5	60.0	1800	380	380	380	86	84	84	57	80	4.42	26.5
49%	5	60.0	1800	380	380	380	170	168	167	112	80	4.30	26.5
75%	5	60.0	1800	380	380	381	260	259	259	172	86	4.16	26.5
99%	10	60.0	1800	380	381	381	343	343	342	226	90	4.08	26.5
110%	10	60.0	1800	380	381	381	379	378	378	250	94	4.00	26.5

LOAD ACCEPTANCE TEST:

SUDDENLY BLOCK LOAD	VOLTAGE	FREQUENCY	CURRENT	RESPONSE TIME
%	V	Hz	A	SEC
49%	380	60.0	170	0
0%	390 – 380	61.6 – 60.0	0	5
50%	371 – 380	58.4 – 60.0	176	5

PROTECTION SYSTEM TEST:

OVER FREQUENCY SD	UNDER FREQUENCY SD	OVER VOLTAGE SD	UNDER VOLTAGE SD	OIL PRESSURE SD	COOLANT LEVEL SD	COOLANT TEMP SD
67	53	460	340	1.0	OK	ECM 105

ACCESSORIES DETAILS :				
	Model		Serial No	
	Model		Serial No	
	Model		Serial No	

Test Carried out by PI Supervisor :		Witnessed and approved by Client :	
Name :	Eng. FARHAN SALEEM	Name :	
Signature :		Signature :	
Date :	20 / JULY / 2017	Date :	

PRECISION INDUSTRIES TESTED Testing Department

54 - 345 - 300

PO BOX NO. 17, BARNACK ROAD, STAMFORD LINCOLNSHIRE, PE9, 2NB, ENGLAND.

STAMFORD

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N17D166242**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCDI274K1	**AVR:** SX460	**AVR NO:** 1115052029	**ENCLOSURE:** IP23	
RATED KVA: 250	**RATED VOLTS:** 400	**RATED AMPS:** 360.8	**FREQUENCY:** 50	**PHASE:** 3
POWER FACTOR: .8	**STR CONN:** Series Star	**STR WINDING:** 311	**DUTY:** Base Continuous	**WIRE:** 4-Wire
EXC. VOLTS: 79	**EXC.AMPS:** 2.9	**AMB TEMP:** 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.0267	2+	OK	H
ROTOR	2.164	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12.3	.51	0.00
50	400	360.8	250	.8	200	79	2.9	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UCD27.4.K.1.311.,1,.,14,.SX460.Black - RAL9011
Remarks:

Date: 21-APR-17	Job No: 3481897-008	AUTHORISED BY **Rameshwar B Karkhande** Quality Assurance

Certificate Of Conformity

Quality System Approval
ISO 9001/2008

Frame Size : **UCDI274K1**

Serial Number : **N17D166242** **Manufactured Date :** 21-APR-17

KVA: **250**

Authorized By Quality Assurance: **Rameshwar B Karkhande**

This document is electronically generated **Print Date: 21-APR-17**

Power Generation

Cummins India Ltd.

Power Generation Business Unit

Engine Test Certificate

This is to certify that Engine Model QSL9-G3

Serial Number 84507971 was tested using

Diesel as per ISO - 3046 Standard.

Corrected Full Load B.H.P. to N.T.P. conditions as per the

test standard is 345 B.H.P. @ 1500 R.P.M.

399BHP @ 1800RPM

Date : 09-Jun-17

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Product Conformance

A048U279

EU DECLARATION OF CONFORMITY

Cummins Generator Technologies

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU	Low Voltage Directive
2014/30/EU	The Electromagnetic Compatibility (EMC) Directive

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005	Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments
EN 61000-6-4:2007+A1:2011	Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments
EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction
EN 60034-1:2010	Rotating electrical machines - Part 1: Rating and performance
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets
BS 5000-3:2006	Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed: Date: 01st February 2016	Name, Title and Address: Kevan J Simon Global Technical and Quality Director Cummins Generator Technologies Fountain Court Lynch Wood Peterborough, UK PE2 6FZ

Description UCDI274K1

Serial Number N17D166242

450-16383-F

Registered in England under Registration No. 441273.
Cummins Generator Technologies Ltd. Registered Office: Barnack Road, Stamford, Lincolnshire PE9 2NB, England.